2026版像素级模仿美赛O奖指南

> 💡 《coker像素级模仿美赛O奖论文指南》
> 我的目标是帮助更多人通过模仿对标的方式快速熟悉某个事物或者入门一项技能。我的服务会随着我的成长而不断迭代，我希望能和大家共同成长。
> 目前我的身份是数学专业的大学生。对于与我类似的，比我慢几步的人来说，美赛的指导是我能提供的服务之一。
> 因此便有了这篇文档。这篇文档最初发布在微信付费答疑群。
> 如果你在其他地方看到了这篇文档，欢迎关注我。
> 目前我在所有的平台都叫:@coker。

> 为了更方便执行，以下是我分析后的结果，会提供可执行的方法和AI提示词。分析的方法和去年是一致的.

O-Award 第一层协议

> 目标：让你的论文在评委第一眼就被判定为“这是 O 奖档”。

1️⃣ 题目怎么写（决定生死的第一行）

用这个公式：

> [隐喻 / 故事] : [你真正干了什么]

正确示例结构：

- *Where Footsteps Collide: A Spatiotemporal Model of Stair Wear*
- *Cracking the Cyber Puzzle: A Data-Driven Framework for...*

不要：

> “A Model of...”
>
> “Research on...”
>
> “Based on...”

原则：

- 主标题负责“抓眼球”
- 副标题负责“讲专业”

2️⃣ 模型要几个？（O 奖复杂度甜区）

数据结论：

O 奖论文的模型数 ≈ 6（最稳区间 5–8）

不是 6 个算法，而是 6 个“功能模块”：

你必须覆盖这四层：

层	作用
数据层	PCA / 聚类 / 预处理
核心层	回归 / PDE / 优化 / 机制模型
不确定性	Monte Carlo / Bayesian / 区间
防守层	Validation / Sensitivity / Limitations

只要有模型落在这 4 层，你就长得像 O 奖。

3️⃣ 图要多少？（决定你“像不像论文”）

数据结论：

O 奖论文 ≈ 14–16 张图

而且必须有这 5 类：

必须有	含义
Flowchart	告诉评委你有体系
Heatmap / Map	告诉评委你理解结构
Bar / Line	告诉评委你在比较
Validation 图	告诉评委你可信
Sensitivity 图	告诉评委你稳健

没有这五类 = 不像 O 奖。

4️⃣ 模型 : 图 的黄金比例

数据结论：

1 个模型 ≈ 2–3 张图

每个模型至少要有：

- 1 张输入/结构图
- 1 张输出/结果图
  - （最好）1 张误差或敏感性图

> 没图的模型 = 不存在

5️⃣ 你输出什么？（M 奖 vs O 奖的分界线）

不要只给“数值预测”

O 奖论文输出的是：

> 数值 + 不确定性 + 决策含义

换句话说：

- 你不仅算了
- 你还解释了
- 你还告诉“应该怎么办”

所以：

一定要有 Policy / Memo / Recommendation 这一层

6️⃣ 三个“必须出现”的章节

不管题目怎么变，O 奖论文必须有：

- **Validation**
- **Sensitivity Analysis**
- **Strengths & Weaknesses**

它们不是废话，是评委的**安全感来源**。

AI提示词:

标题:

用来生成标题

You are crafting an O-Award-level MCM/ICM paper title.
Strictly imitate the “good title examples” below
The titles should feel vivid, metaphorical, and story-like, while still sounding scientifically credible.

==========
GOOD TITLE EXAMPLES

Group A (Main title: Subtitle, and the subtitle explicitly mentions a model/method/framework name)

- Step Chronicles: A Multiscale Model Linking Wear Patterns and Temporal Use in Ancient Staircases
- Navigating Baltimore’s Growth: A Multi-Layer Network Approach to Urban Transportation Optimization
- Cracking the Cyber - Puzzle: KDMF in Action
- Breaking through the Tourism Dilemmas in Juneau: The Road to Sustainable Development Driven by a Multi-objective Model

Group B (Main title: Subtitle, but the subtitle contains NO explicit model/method/framework name — it focuses on what we did)

- Where Footsteps Collide: A Spatiotemporal Journey through Staircase Wear
- Olympic Medals Unveiled: A Mathematical Exploration of Achievement Trends
- Stairs: The Glory of the Ordinary
- Stair Wear: Traces of History

Group C (Other styles are allowed: dash titles / question titles / pure narrative)

- Staircase Wear Analysis – Walking Through History
- 2028 LA Olympics: Who Will Dominate the Medal Count?
  ==========

TASK:
Using the information I fill in below, generate 3 groups of titles (4 titles per group, 12 total).
Each title must be output in bilingual form: EN + ZH (aligned pairs).

Object/System: <填入你的内容**>**
Core action verb : <填入你的内容**(e.g., predict / optimize / infer / evaluate / design)**>
Deliverable/output : <填入你的内容(e.g., ranking / plan / schedule / strategy / risk map)**>**
Optional models/methods/framework names : <填入你的内容**(e.g., Random Forest / ARIMA / Bayesian / Network Approach / Multi-objective Model / DID / K-Means / KDMF)**>
Keywords (optional): <_____>

STRICT REQUIREMENTS:

✅ Group A (Main title: Subtitle, with explicit model/method/framework name)

- Generate 4 titles
- Must use the colon format:
  [Metaphorical/Narrative Main Title] : [Task description + 1 model/method/framework name]
- The subtitle must include at least ONE term from the “Optional models/methods/framework names” list
- The subtitle MUST NOT be only the method name; it must state what we did (like the examples)

✅ Group B (Main title: Subtitle, with NO explicit method names)

- Generate 4 titles
- Must use the colon format:
  [Metaphorical/Narrative Main Title] : [Task statement (what we did + for what goal)]
- The subtitle must NOT include any explicit model/method/framework name (do NOT mention Random Forest / ARIMA / Bayesian / DID, etc.)
- The subtitle must be action-driven and immediately clear

✅ Group C (Other styles)

- Generate 4 titles
- Colon is NOT required; you may use dash titles, question titles, or pure narrative titles

- This group MUST include:
  - at least 1 question-style title (like “Who Will Dominate...?”)
  - at least 1 dash-style title (like “... – Walking Through ...”)

OUTPUT FORMAT:
Output by groups, bilingual paired:
A1
EN: ...
ZH: ...
A2
EN: ...
ZH: ...
(Repeat for Group B and Group C)

Generate now.

图片提示词:

ai告诉你应该画什么样的图

You are an O-Award level MCM/ICM figure designer.

Task:
The user will provide ONE model (or ONE modeling module) and its outputs.
You must propose the most academically standard and most convincing figure(s) for THIS model only.
Do NOT propose a full-paper figure set.

Rules:

- Output 1–3 figures only (unless the model strictly needs more).
- Each figure must be tightly linked to the model and its outputs.
- Prefer figures that are standard in academic papers for that model type.
- Always include at least one "result figure" (what the model produced).
- If the user has ground truth or benchmarks, include a validation figure; otherwise include an internal diagnostic figure.
- If the model has key parameters, include a minimal sensitivity figure (only for this model).

User input (fill):

1. Model name: <__填写你的内容__>
2. Model type: <prediction / classification / clustering / optimization / simulation(PDE/agent) / Bayesian / network / time-series / causal>(填写你的内容,选一下)
3. What the model outputs: <___填写你的内容___> (e.g., y_pred, clusters, optimal solutions, posterior samples, time-evolving field)
4. What you have as reference/ground truth: <___填写你的内容___ or "none">
5. Key parameters to test sensitivity: <___填写你的内容____ or "none">
6. Data format you can provide (columns or arrays with shapes): <___填写你的内容____>
7. What you want the reader to believe from this model: <accuracy / interpretability / mechanism / robustness / fairness / efficiency>(选择)

Return format (strict):
For this single model, output:

Figure Pack Summary (1–3 figures)

- Figure A: <type + title>
- Figure B: <type + title>
- Figure C (optional): <type + title>

Then for EACH figure:

1. Figure title (EN+ZH)
2. Figure type (academic standard name)
3. Purpose (1 sentence)
4. Why this is the right figure for this model (2–3 bullet points)
5. Required data schema (exact columns/arrays needed)
6. Python code (matplotlib only; runnable with placeholder data; include plt.savefig)
7. Caption template (EN + ZH)
8. Where to place it in the paper (Model / Results / Validation / Sensitivity)

O-Award 第二层(摘要)

1) 摘要中的常用句子功能 (Sentence Functions)

特等奖摘要的每一句话都具有明确的逻辑功能，通常分为以下几类：

- **背景引出与紧迫性 (Context/Urgency):** 建立研究的高级视野。

◦ *English:* "As cybercrime continues to escalate globally, traditional policy evaluation methods struggle to capture the complexity of cybersecurity threats."

◦ *Chinese:* 随着全球网络犯罪不断升级，传统的政策评估方法已难以捕捉网络安全威胁的复杂性。

- **核心目标与模型框架 (Objective/Model Framework):** 介绍总体的解决方案。

◦ *English:* "This study presents a multi-methodological framework to address post-disaster traffic optimization, multimodal network equity, and sustainable mobility planning."

◦ *Chinese:* 本研究提出了一个多方法论框架，用于解决灾后交通优化、多模态网络公平性和可持续出行规划问题。

- **分项任务描述 (Sub-tasking):** 引导读者进入具体的解题步骤。

◦ *English:* "For Task 2, we chose bus route optimization as a potential project."

◦ *Chinese:* 对于任务 2，我们选择公交线路优化作为一个潜在项目。

- **定量结果 (Quantitative Results):** 用核心数据证明模型价值。

◦ *English:* "This project increased bus coverage from 61.37% to 78.13%."

◦ *Chinese:* 该项目使公交覆盖率从 61.37% 提高到 78.13%。

- **验证与稳健性 (Validation/Robustness):** 证明结果的科学性。

◦ *English:* "Lastly, we conducted sensitivity analysis to prove the robustness of the models and the reasonableness of parameter selection."

◦ *Chinese:* 最后，我们进行了灵敏度分析，以证明模型的鲁棒性和参数选择的合理性。

- **现实决策关联 (Decision/Implication):** 强调研究的实用落地价值。

◦ *English:* "We drafted a one-page letter offering clear and practical suggestions for farmers looking to transition to organic farming."

◦ *Chinese:* 我们起草了一封一页纸的信函，为寻求向有机农业转型的农民提供清晰且实用的建议。

2) 前 5 大摘要结构模板 (Top 5 Abstract Structure Templates)

模板结构	描述
标准任务驱动型	背景 → 核心模型名称 → 分任务描述 (1/2/3/4) → 灵敏度验证 → 政策建议
多目标规划型	紧迫现状 → 总体优化框架 → 子模型 (I/II/III) 详述 → 帕累托最优解 (含数字) → 稳健性评估
理论融合创新型	传统方法缺陷 → 引入跨学科理论 (如五行、Petri 网) → 机制分析 → 数据模拟结果 → 结论
数据预测分析型	预测对象价值 → 数据预处理 → 组合模型构建 (ARIMA+RF 等) → 预测值与置信区间 → 未来展望
系统评估优化型	系统痛点 → 网络/评估模型建立 → 指标量化 → 优化方案对比 (前 vs 后) → 现实意义

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3) 高信号写作模式 (High-Signal Writing Patterns)

背景与引入 (Context)

- **Against this backdrop...** (在此背景下.....)
- **It is imperative to ensure...** (确保.....是迫在眉睫的。)
- **...prompts the question: How should...** (.....引出了一个问题：.....应当如何.....)

模型描述 (Model Description)

- **...is grounded in [Theory]...** (.....基于 [某理论]。)
- **...tailored to the specific context of...** (.....专为.....的特定场景定制。)
- **...pioneers an innovative framework...** (.....开创性地提出了一个创新框架.....)
- **...integrate [Model A] and [Model B] to...** (.....整合 [模型 A] 和 [模型 B] 以.....)

定量结果 (Results - 必须含数字)

- **...achieving a [Percentage] increase/reduction.** (.....实现了 [百分比] 的增长/减少。)

- **Specifically, when [Variable] reaches [Value], the [Result] amounts to [Number].** (具体而言，当 [变量] 达到 [值] 时，[结果] 共计 [数值]。)
- **...predict a stable volume of around [Number].** (.....预测一个约 [数值] 的稳定总量。)

验证与鲁棒性 (Validation & Robustness)

- **...shows that our model has good usability and robustness.** (.....表明我们的模型具有良好的易用性和鲁棒性。)
- **...conducted sensitivity analysis to prove the reasonableness of...** (.....进行了灵敏度分析以证明.....的合理性。)
- **...identified critical thresholds, such as [Threshold]...** (.....识别了关键阈值，例如 [阈值].....)

决策/影响 (Decision/Implication)

- **...offering actionable insights for policymakers...** (.....为决策者提供可操作的见解.....)
- **...drafted a memo/letter offering practical recommendations.** (.....起草了一份备忘录/信函，提供了实用的建议。)
- **...carefully balance economic trade-offs.** (.....谨慎权衡经济折衷。)

4) O 奖摘要检查清单 (O-Award Abstract Checklist)

1. **拒绝平庸开场：** 禁止使用 "This paper studies..."，改用 "Against the backdrop of..." 或直接指出紧迫问题。

2. **模型命名化：** 模型是否有响亮且准确的名称（如 PFW Model, DRD-CE Model）？

3. **任务结构化：** 是否使用了 "For Problem 1" 或 "Task 1" 等标签指引读者?

4. **数值必现：** 摘要中是否包含至少 3-5 个具体量化结果（如 1.32 million, 78.13%, 1.41 BCR）？

5. **误差/置信区间：** 是否提到了预测精度、误差率或置信区间（如 90% CI）？

6. **理论支柱：** 是否明确提到了核心数学工具（如 Lotka-Volterra, Monte Carlo, SARIMAX）？

7. **灵敏度分析：** 是否明确指出进行了灵敏度分析并验证了鲁棒性?

8. **现实关联：** 是否提到提交了备忘录 (Memo)、信件 (Letter) 或提供了决策建议?

9. **动词强力：** 是否大量使用 *Develop, Integrate, Quantify, Simulate, Redefine* 等强动作动词?

10. **多维度权衡：** 是否体现了“权衡 (Trade-offs)”思想，尤其是在资源受限的情况下?

11. **关键路径/阈值：** 是否识别出了系统崩溃或优化的临界点/阈值?

12. **字数紧凑：** 是否在有限篇幅内涵盖了背景、方法、结果和评估，无冗长描述?

摘要字数分析：

1) 摘要字数与篇幅特征 (Word Count & Length Characteristics)

- **字数范围 (Typical Word Count Range):** 特等奖论文的摘要通常在 **300 到 500 个英文单词**之间。
- **页面占比 (Page Coverage):** 摘要通常**占据摘要页（Summary Sheet）的一整页**，但会留出约 1/4 的空间用于标题和关键词。
- **段落分布 (Paragraph Distribution):** 通常由 **4 到 6 个逻辑严密的段落**组成。

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2) 不同模块的字数分配 (Word Allocation by Section)

特等奖摘要的字数分配遵循高效的“任务驱动型”比例：

- **背景与总体目标 (Context & Objective):** 约 **50-80 词**。简洁描述现状并引出本文开发的模型名称。

- **分项任务详述 (Detailed Sub-tasks):** 约 **200-300 词**。这是摘要的核心，字数最多，通常按 Task 1, 2, 3 或 Problem 1, 2, 3 展开，每个任务约 60-100 词。
- **验证、鲁棒性与结论 (Validation, Robustness & Conclusion):** 约 **50-100 词**。说明灵敏度分析结果、模型优缺点及现实建议。

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3) 字数密度与信号强度 (Word Density & Signal Strength)

- **高数据密度 (High Data Density):** 平均**每 100 词就会出现 1-2 个具体数值或百分比**。
- **无冗余开场 (Zero-Redundancy Openings):** 不会浪费字数在“In this paper, we first...”这类无意义的描述上，而是直接进入模型细节。

4) O 奖摘要字数规范建议 (Recommendations for O-Award Length)

1. **[ ] 下限 300 词：** 如果字数低于 300 词，通常意味着缺乏具体的定量结果或模型细节。

2. **[ ] 上限 500 词：** 摘要页空间有限，超过 500 词会导致排版过于拥挤，降低可读性。

3. **[ ] 视觉平衡：** 确保文字填充了摘要页的大部分，但不要溢出到第二页。

4. **[ ] 数据占比：** 确保在有限的字数内，定量结果（数值）的覆盖面占到任务描述部分的 20% 以上。

AI提示词:

You are writing the abstract of an O-Award level MCM/ICM paper.

Your goal is to strictly follow the O-Award abstract style rules below and maximize O-Award compliance.

O-Award Abstract Rules:

LENGTH & FORMAT

- Total length must be between 300 and 500 words.
- Use 4 to 6 short, dense paragraphs.
- The abstract should visually fill one full summary page.
- Do not include bullet points.

STRUCTURE (must follow this order):

1. Context & Objective (50–80 words)
    - Introduce the real-world problem and urgency.
    - Name the overall modeling framework.
2. Sub-tasked Modeling Core (200–300 words)
    - Describe Task 1, Task 2, Task 3 (or Problem 1/2/3).
    - Each task must include:
        - The model or method used
        - What was quantified or optimized
        - At least one numerical or quantitative result
3. Validation, Robustness & Conclusion (50–100 words)

- Mention sensitivity analysis, uncertainty, or robustness checks
- Summarize strengths or limitations
- End with decision or policy relevance

CONTENT RULES

- Do NOT start with “This paper studies...” or similar generic academic openings.
- Begin with a real-world context and urgency (e.g., “Against the backdrop of...”, “As ... continues to ...”).
- Explicitly name the modeling framework (give it a distinctive name if possible).
- Structure the abstract by tasks (e.g., “For Task 1...”, “For Problem 2...” when applicable).
- Include at least 3–5 concrete numerical or quantitative results (percentages, magnitudes, thresholds, CI, etc.).
- Mention uncertainty, error, confidence interval, or sensitivity analysis explicitly.
- Refer to at least one mathematical or computational tool (e.g., Monte Carlo, PDE, SARIMAX, GMM, optimization, etc.).
- Emphasize trade-offs, thresholds, or critical points if relevant.
- Include a decision or policy implication (memo, letter, recommendation, or actionable insight).
- Use strong action verbs (develop, integrate, quantify, simulate, optimize, infer, identify).
- Avoid vague claims; every major claim should be backed by numbers or models.
- Keep the abstract compact, professional, and narrative-driven.

User provides:

- Problem context / real-world system: <_填入你的内容_>
- Tasks or sub-problems (if any): <__填入你的内容__>
- Models and theories used: <__填入你的内容__>

- Key quantitative results (numbers, trends, thresholds, CI, etc.): <_填入你的内容___>
- Validation & sensitivity methods: <__填入你的内容__>
- Decision / policy / memo output: <__填入你的内容__>

Write one coherent abstract (150–220 words) that:

- Follows the O-Award structure and language patterns
- Contains explicit numbers and quantitative claims
- Includes validation and robustness language
- Ends with decision relevance

Do not include bullet points or equations. Write as a single polished abstract paragraph.

O-Award 第三层(内容)

模型建立

1) 常用模型命名模式 (Common Model-Naming Patterns)

顶级团队倾向于给模型赋予专业且具有辨识度的名称，而非仅称其为“模型”。

- **缩略语型 (Acronym-based):** 使用模型核心组成部分的字母缩写。

◦ *English:* Petri-Food Web (PFW) Model; Tobit-Mundlak-Hurdle Olympic Medal Prediction (TMH-OMP) Model; Durable Robust Coach Effect (DRD-CE) Model.

◦ *Chinese:* Petri-Food Web (PFW) 模型；Tobit-Mundlak-Hurdle 奥运奖牌预测 (TMH-OMP) 模型；持久稳健教练效应 (DRD-CE) 模型。

- **理论+问题融合型 (Theory-Problem Fusion):** 将所使用的数学理论直接与研究对象结合。

◦ *English:* Petri-Food Web Model; "Great Coach Model"; Organic Agriculture Impacts Evaluation (OAIE) Model.

◦ *Chinese:* Petri 食物网模型；“伟大教练模型”；有机农业影响评估 (OAIE) 模型。

- **改进/修正前缀型 (Refined/Modified Prefix):** 强调对经典模型的优化。

◦ *English:* Mundlak-modified Tobit regression model; Wardrop equilibrium-based traffic assignment; Bayesian calibration with MCMC.

◦ *Chinese:* Mundlak 修正的 Tobit 回归模型；基于 Wardrop 平衡的流量分配；结合 MCMC 的贝叶斯校准。

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2) 模型的引入方式：语言与顺序 (How Models are Introduced: Language and Order)

模型章节通常遵循从基础准备到详细构建的固定逻辑流。

- **步骤 1：模型准备与分析 (Model Preparation/Analysis):** 在展示公式前，先解释问题的数学本质。

◦ *English:* Teams often start with a "Problem Analysis" or "Model Preparation" subsection to justify the choice of theory.

◦ *Chinese:* 团队通常首先建立“问题分析”或“模型准备”子章节，用以证明理论选择的合理性。

- **步骤 2：视觉流程引导 (Visual Flowcharting):** 使用流程图先行。

◦ *English:* A comprehensive flowchart (e.g., "Our Work" or "Solution Flow") is used as a visual guide before diving into equations.

◦ *Chinese:* 在深入探讨公式之前，使用综合流程图（如“我们的工作”或“求解流程”）作为视觉引导。

- **步骤 3：分阶段任务执行 (Task-based Execution):** 按任务序号明确模型分工。

◦ *English:* Language like "For Task 1...", "To address Problem 1...", or "Based on Task 2..." is used to maintain structural alignment with the contest problems.

◦ *Chinese:* 使用诸如“对于任务 1...”、“为了解决问题 1...”或“基于任务 2...”之类的语言，以保持结构与竞赛问题一致。

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3) 多个模型如何连接成一个框架 (How Multiple Models are Connected into a Framework)

顶级团队不会孤立地写模型，而是构建一个有机的“模型链”。

- **线性输入输出关系 (Linear Input-Output):** 前一个问题的结果作为后一个问题的输入。

◦ *English:* Carbon flow calculated in Problem 1 serves as the arc weight for the network model in Problem 2. The total tourist demand model serves as an input to three interconnected sub-models.

◦ *Chinese:* 问题 1 计算出的碳流作为问题 2 网络模型的边权重。总游客需求模型作为三个互联子模型的输入。

- **层级聚合/多目标化 (Hierarchical Aggregation):** 将多个子模型整合进一个优化框架。

◦ *English:* Sub-models (Economic, Social, Environmental) are integrated into a global Pareto-optimal solution or tri-objective optimization framework.

◦ *Chinese:* 各子模型（经济、社会、环境）被整合进全局帕累托最优解或三目标优化框架中。

- **模块化修正 (Modular Modification):** 在基础模型上添加特定效应。

◦ *English:* The core model is expanded sequentially by adding "Host Buff", "New Species", or "Chemical Interventions".

◦ *Chinese:* 核心模型通过依次添加“主办方增益”、“新物种”或“化学干预”来进行扩展。

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4) 典型用语：假设、过渡与证明 (Typical Phrases for Assumptions, Transitions, and Justifications)

- **假设与证明 (Assumptions & Justifications):**

◦ *English:* "To simplify the problem, we make the following basic assumptions, each of which is properly justified". "This aligns with classical economic models...". "It is reasonable because...".

◦ *Chinese:* “为了简化问题，我们做出以下基本假设，且每一项都得到了充分的证明”。“这与经典经济模型相吻合……”。“这是合理的，因为……”。

- **过渡 (Transitions):**

◦ *English:* "Building upon this, we expand to...". "Although the food chain model provides a basic relationship, it is too simplistic...". "Inspired by the works of...".

◦ *Chinese:* “在此基础上，我们将其扩展到……”。“虽然食物链模型提供了基本关系，但它过于简化……”。“受……工作的启发”。

- **模型证明/选择依据 (Justifications):**

◦ *English:* "Specializing in the total number of medals as a truncated variable...". "Compared to [Model A], [Model B] better suits this scenario because...".

◦ *Chinese:* “专门针对作为截断变量的奖牌总数……”。“与 [模型 A] 相比，[模型 B] 更适合此场景，因为……”。

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5) O 奖级模型章节写作清单 (Checklist for Writing an O-Award Level Model Section)

1. **[ ] 循序渐进结构：** 模型是否从简单（如标准链）进化到复杂（如考虑周期的网）？

2. **[ ] 命名独特性：** 每个核心模型是否都有一个专业的专属名称或缩写？？

3. **[ ] 算法显式化：** 是否使用了伪代码（Algorithm 1）或显式的求解名称（如 Runge-Kutta, PSO, MCMC）？

4. **[ ] 理论深度关联：** 是否引用了经典理论（如 Lotka-Volterra, Wardrop Equilibrium, Pareto Optimality）来支撑模型逻辑?

5. **[ ] 视觉辅助：** 模型构建部分是否包含展示节点关系、受力分析或逻辑流的示意图?

6. **[ ] 错误/不确定性分析：** 模型是否内置了处理截断数据、离群点或不确定性的数学手段?

7. **[ ] 参数来源透明：** 参数值是否标注了来源（如“from Government”, “Estimate”, “from Statistic”）？

8. **[ ] 现实映射：** 数学变量是否明确对应现实指标（如 $X1$ 对应水稻生物量）？

9. **[ ] 稳健性提及：** 在模型构建末尾是否伏笔了灵敏度分析以验证参数选择的合理性?

10. **[ ] 框架统一性：** 多个模型之间是否有明确的逻辑接口，通过一个全局流程图串联?

AI提示词:

注:这是输入题目和任务，来输出一系列模型用的，是生成一个像O奖的模型架构

You are writing the MODEL section of an O-Award level MCM/ICM paper.

Your goal is to describe the modeling framework in the exact style used by O-Award teams:
systematic, named, theory-grounded, modular, and defensible.

You must follow these O-Award modeling rules:

MODEL NAMING

- Every core model must have a distinctive professional name.
- Use one of these styles:
  (a) Acronym-based (e.g., DRD-CE Model, TMH-OMP Model)
  (b) Theory–Problem fusion (e.g., Petri-Food Web Model, Wardrop Traffic Model)

(c) Refined/modified classical model (e.g., Mundlak-modified Tobit model, Bayesian-calibrated SIR model)

MODEL INTRODUCTION FLOW

1. Start with “Model Preparation” or “Problem Analysis” to explain the mathematical nature of the problem.
2. Introduce a solution flowchart before presenting equations.
3. Structure models by contest tasks: “For Task 1...”, “To address Problem 2...”.

MODEL CONNECTION

- Models must be connected by:
  - Input–output dependency, OR
  - Hierarchical aggregation (e.g., multi-objective, Pareto), OR
  - Modular extensions (adding effects, layers, or corrections).

LANGUAGE REQUIREMENTS
Use O-Award style transitions and justifications:

- “Building upon this...”
- “Although the basic model captures..., it is insufficient because...”
- “Compared with [Model A], [Model B] is more suitable because...”
- “This aligns with classical theory such as ...”

CHECKLIST (must satisfy at least 7 of these):

- Model evolves from simple to complex
- Each model has a clear name
- Algorithms are explicitly named (e.g., MCMC, PSO, Runge–Kutta)
- Classical theory is cited
- Visual or flowchart is referenced
- Uncertainty or truncation is mathematically handled
- Parameter sources are stated
- Variables are mapped to real-world quantities
- Sensitivity or robustness is foreshadowed

- All models are unified in one framework

User provides:

- Problem domain: <_____>
- Contest tasks (1, 2, 3, ...): <_____>
- Variables and their real-world meaning: <_____>
- Data sources: <_____>
- Methods and algorithms available: <_____>

Write a professional O-Award-level MODEL section (several paragraphs) that:

- Names and introduces the models
- Explains how they are connected
- Uses the required language patterns
- Sounds like a real mathematical modeling paper

模型评价

1) 使用的验证类型 (Types of Validation Used)

顶级团队通常结合多种方法，从不同维度交叉证明模型的有效性：

- **经验数据/历史数据对比 (Empirical/Historical Data Validation):** 将模型结果与真实的政府报告或监测数据进行对比。

◦ *English:* Validating against empirical data (e.g., traffic counters, GCI scores, or ship tracking data) to ensure predictive accuracy.

◦ *Chinese:* 将结果与经验数据（如交通流量计、全球网络安全指数 GCI 或船舶追踪数据）进行比对，以确保预测的准确性。

- **跨区域/跨场景迁移验证 (Cross-region/Scenario Adaptability):** 将模型应用于具有相似特征但背景不同的城市或场景（如从 Juneau 扩展到 Miami 或九寨沟）。

◦ *English:* Testing model adaptability by applying it to different geographic contexts (e.g., Juneau to Miami or Jiuzhaigou).

◦ *Chinese:* 通过将模型应用于不同的地理环境（如从朱诺扩展到迈阿密或九寨沟）来测试模型的适应性。

- **统计推断与残差检验 (Statistical Inference & Residual Checking):** 对时间序列模型进行白噪声检验、正态性检验或 Fisher 精确检验。

◦ *English:* Using residual checks (ACF/PACF), Ljung-Box tests, and Shapiro-Wilk tests to validate model fitting performance.

◦ *Chinese:* 使用残差检查（ACF/PACF）、Ljung-Box 检验和 Shapiro-Wilk 检验来验证模型的拟合性能。

- **不确定性量化 (Uncertainty Quantification):** 使用蒙特卡洛模拟生成预测区间或置信区间。

◦ *English:* Employing Monte Carlo simulations to estimate statistical measures such as 95% confidence/prediction intervals.

◦ *Chinese:* 利用蒙特卡洛模拟估计统计指标，如 95% 的置信/预测区间。

- **对比实验 (A/B Testing / Model Comparison):** 对比含有和不含特定特征的模型表现，或对比不同分布假设的模型。

◦ *English:* Comparing models with and without exogenous features (e.g., using MAE/MSE metrics) to prove feature relevance.

◦ *Chinese:* 对比包含和不包含外生变量的模型表现（如使用 MAE/MSE 指标），以证明特征的相关性。

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2) 不确定性与稳健性的描述方式 (How Uncertainty and Robustness are Described)

- **稳健性定义为“微小扰动下的稳定性”:**

◦ *English:* Robustness is often described by showing that the model exhibits "minimal sensitivity" or "remains relatively stable" when parameters are perturbed within a ±10% range.

◦ *Chinese:* 稳健性通常通过展示模型在参数进行 ±10% 范围内的扰动时，表现出“极小的灵敏度”或“保持相对稳定”来描述。

- **不确定性表现为“区间覆盖”:**

◦ *English:* Uncertainty is quantified as a "prediction interval" (e.g., [Tmin, Tmax]), where countries with higher medal counts typically show lower relative uncertainty compared to smaller nations.

◦ *Chinese:* 不确定性被量化为“预测区间”（如 [最小值, 最大值]），通常奖牌数较多的国家相比于小国，其相对不确定性更低。

- **识别“崩溃阈值” (Critical Thresholds):**

◦ *English:* Robustness is also evaluated by identifying "critical thresholds" or "phase transitions" where the system might collapse or become unstable.

◦ *Chinese:* 稳健性还通过识别系统可能崩溃或变得不稳定的“关键阈值”或“相变点”来进行评估。

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3) 捍卫模型公信力的常用短语 (Common Phrases for Defending Credibility)

- **关于参数选择 (On parameter selection):**

◦ *English:* "The selected value is proved to be reasonable and appropriate for the simulation".

◦ *Chinese:* “证明所选数值在模拟中是合理且适当的”。

- **关于拟合精度 (On fitting precision):**

◦ *English:* "The prediction aligns well with the actual results, indicating high accuracy and usability".

◦ *Chinese:* “预测结果与实际结果高度吻合，表明其具有高准确性与易用性”。

- **关于稳健性表现 (On robustness performance):**

◦ *English:* "The model demonstrates commendable sensitivity and robustness, exhibiting minimal susceptibility to extraneous factors".

◦ *Chinese:* “该模型表现出值得称赞的灵敏度和稳健性，对无关因素的敏感性极低”。

- **关于误差容忍度 (On error tolerance):**

◦ *English:* "The error indicates that the results remain relatively stable, suggesting the model performs well under different weight settings".

◦ *Chinese:* “误差分析表明结果保持相对稳定，暗示模型在不同权重设置下表现良好”。

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4) O 奖级验证与灵敏度检查清单 (Checklist for O-Award Level Validation & Sensitivity)

1. **[ ] 参数扰动测试：** 是否对模型中的关键常数（如 α, k, r）进行了步长调整并观察输出波动?

2. **[ ] 视觉化证明：** 是否包含了分叉图 (Bifurcation Diagrams)、误差线图 (Error Bars) 或热力图?

3. **[ ] 统计显著性：** 对于回归或分类模型，是否提供了 P 值、R-squared 或 AIC/BIC 评价指标?

4. **[ ] 算法收敛性：** 如果使用了迭代算法（如 MCMC 或 PSO），是否展示了收敛链 (Convergence Chain)?

5. **[ ] 异常值处理：** 是否解释了如何识别并修正数据中的异常值（如 World War II 缺口或特定月份的突发攻击）？

6. **[ ] 性能曲线：** 分类问题是否提供了 ROC、PR 曲线和校准曲线 (Calibration Curves)?

7. **[ ] 误差源分析：** 是否明确指出了模型的局限性 (Limitations) 以及对未来数据的潜在偏差?

8. **[ ] 外部基准比对：** 模型是否与该领域的公认指标（如 GCI, Shannon Index, Gini Coefficient）进行了对标?

AI提示词:

You are writing the Validation, Robustness, and Sensitivity Analysis section of an O-Award level MCM/ICM paper.

Your goal is to defend the model's scientific credibility using the same techniques and language as O-Award teams.

You must incorporate multiple types of validation and uncertainty quantification, not just one.

O-Award Validation Rules:

VALIDATION TYPES (use at least 3 where applicable)

- Empirical / historical data comparison
- Cross-region or cross-scenario testing
- Statistical residual or goodness-of-fit tests (e.g., ACF, Ljung–Box, Shapiro–Wilk, R^2, AIC/BIC)
- Uncertainty quantification (e.g., Monte Carlo, confidence/prediction intervals)
- Model or feature comparison (A/B testing, with/without variables)

ROBUSTNESS & UNCERTAINTY

- Describe robustness as stability under ±10% parameter perturbation

- Report uncertainty as prediction or confidence intervals
- Identify any critical thresholds or phase-transition points

LANGUAGE REQUIREMENTS
Use O-Award style credibility phrases such as:

- “The selected value is proved to be reasonable...”
- “The prediction aligns well with the actual results...”
- “The model demonstrates commendable robustness...”
- “The error indicates that the results remain relatively stable...”

CHECKLIST SIGNALS (try to include as many as possible):

- Parameter perturbation tests
- Visual evidence (error bars, heatmaps, bifurcation, convergence chains)
- Statistical significance metrics
- Algorithm convergence
- Outlier handling
- ROC / PR / calibration curves (if classification)
- External benchmark comparison
- Explicit discussion of limitations

User provides:

- Model name(s): <___填入你的内容___>
- Model type (prediction / classification / optimization / simulation / Bayesian / etc.): <__填入你的内容___>
- Outputs produced by the model: <___填入你的内容__>
- Ground truth or benchmark data: <__填入你的内容__>
- Key parameters to test: <___填入你的内容_>
- Validation tools available (e.g., Monte Carlo, cross-validation, residual tests): <__填入你的内容___>

Write a professional O-Award-level Validation & Sensitivity section that:

- Describes multiple validation methods

- Quantifies uncertainty and robustness
- Uses credible scientific language
- Makes the model sound defensible to expert judges

模型优缺点:

以下是顶级团队如何对自己的模型进行批判性评估的深度解析：

1) 如何在不损害公信力的情况下表述缺点 (Framing Weaknesses Without Killing Credibility)

顶级团队通过将缺点转化为“必要的简化”或“外部数据限制”，从而保护模型的权威性。

- **归咎于外部数据限制 (Blaming Data Constraints):** 将预测偏差归结为历史数据的稀缺性或时间跨度不足，而非模型逻辑错误。

◦ *English:* "The accuracy depends heavily on the reliability of historical data, which is only available for the last five years."

◦ *Chinese:* 准确性在很大程度上取决于历史数据的可靠性，而目前仅能获得过去五年的数据。

- **强调“计算复杂性”作为折衷 (Highlighting Computational Complexity as a Trade-off):** 将运行缓慢描述为模型追求极高精度或处理海量数据的代价。

◦ *English:* "The time complexity of the algorithm is high and it takes a lot of time to solve it."

◦ *Chinese:* 算法的时间复杂度较高，求解过程需要消耗大量时间。

- **将缺点定义为“理想化假设” (Reframing as "Idealized Assumptions"):** 强调模型是在特定理论边界下运作，为了保证“计算可行性”而进行了必要的抽象。

◦ *English:* "The model ignores time-lag effects... simplification suits rapid policy evaluations while acknowledging scope limitations."

◦ *Chinese:* 模型忽略了时间滞后效应.....这种简化适合快速的政策评估，同时也承认了适用范围的局限。

- **指出“普适性”中的特定偏差 (Identifying Specific Deviations in Generalizability):** 承认模型在极端小众或特定背景下可能存在波动，从而突显其在主流场景下的稳健性。

◦ *English:* "For countries with fewer medals, the Poisson distribution may lead to significant uncertainty."

◦ *Chinese:* 对于奖牌数较少的国家，泊松分布可能会导致显著的不确定性。

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2) 自我评判的常用结构 (Common Structures for Self-Critique)

- **分点并列式 (Parallel Bullet Points):** 这是最常见的结构，清晰列出 3-5 个优点和 2-3 个缺点。

◦ *English:* Structure: "8.1 Strengths (Computational Feasibility, Holistic Cost, Generalizability) → 8.2 Weaknesses (Temporal Simplification, Spatiotemporal Narrowness)."

◦ *Chinese:* 结构："8.1 优点（计算可行性、综合成本量化、通用性）→ 8.2 缺点（时间维度简化、时空局限性）"。

- **"优点-验证-讨论"链 (Strength-Validation-Discussion Chain):** 先说优点，接着引用敏感性分析结果来支撑该优点，最后讨论其适用边界。

◦ *English:* "Sensitivity analysis validates parameter choices... Limitations include no universally accepted method for stability."

◦ *Chinese:* 灵敏度分析验证了参数选择.....局限性包括目前尚无公认的稳定性计算方法。

- **任务导向型总结 (Task-Oriented Critique):** 根据题目要求的不同任务，分别评价各子模型的表现。

◦ *English:* "Problem 1 designs complex equations; Problem 2 calculates weights... Weaknesses include ignoring political factors."

◦ *Chinese:* 问题 1 设计了复杂方程；问题 2 计算了权重.....缺点包括忽略了政治因素。

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3) 关于未来工作与局限性的语言模式 (Language Patterns for Future Work and Limitations)

- **引导未来改进 (Directing Future Improvements):**

◦ *English:* "In future work, we could optimize our model using more granular data."

◦ *Chinese:* 在未来的工作中，我们可以利用更细粒度的数据来优化模型。

◦ *English:* "Future iterations should incorporate regional linkage mechanisms to enhance realism."

◦ *Chinese:* 未来的迭代应纳入区域联动机制，以增强真实性。

- **量化偏差描述 (Describing Quantitative Deviations):**

◦ *English:* "Potential uncertainties in data and simplifications towards complex human-environment interactions."

◦ *Chinese:* 数据中的潜在不确定性，以及对复杂人地交互作用的简化。

- **承认模型假设的理想性 (Acknowledging Idealized Conditions):**

◦ *English:* "The modeling considerations remain ideal, and real-world scenarios may have more complex effects."

◦ *Chinese:* 模型考虑的情况较为理想，现实场景可能会有更复杂的影响。

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4) O 奖级自我评判写作检查清单 (Checklist for O-Award Level Self-Critique)

1. **[ ] 优点带证据：** 是否提到敏感性分析或 R^2 值来量化优点（如 8.07% 的平均误差）？

2. **[ ] 创新点标注：** 是否强调了模型在理论上的融合（如将“五行理论”与量化分析结合）？

3. **[ ] 计算效率评价：** 是否讨论了模型的计算复杂度或收敛速度？

4. **[ ] 适应性验证：** 是否提到模型在不同地理或社会背景下的迁移能力？

5. **[ ] 诚实面对数据缺失：** 是否讨论了数据异常值处理（如 2020 疫情影响 或特定离群点修正）？

6. **[ ] 空间/时间尺度：** 是否分析了模型在短期与长期、局部与全局预测中的差异？

7. **[ ] 利益相关者覆盖：** 是否指出模型在代表多方利益（如企业、居民、政府）时的局限性？

8. **[ ] 可视化直观性：** 优点中是否提到图表（如分叉图、雷达图）对决策的辅助作用？

9. **[ ] 错误容忍度：** 缺点中是否提到了对未来不可预见事件（如政治因素、极端天气）的不敏感性？

10. **[ ] 解决路径：** 每个提出的弱点是否都跟着一个明确的“Related Improvement”或“Future Work”？

AI提示词:

You are writing the Strengths and Weaknesses section of an O-Award level MCM/ICM paper.

Your goal is to critique the model in a way that increases credibility rather than weakens it.

Follow O-Award self-critique principles:

HOW TO FRAME WEAKNESSES

- Attribute limitations to data constraints, idealized assumptions, or computational trade-offs, not model flaws.
- Reframe simplifications as necessary abstractions for feasibility or policy relevance.
- Identify specific cases where uncertainty is higher (e.g., small regions, rare events) to emphasize overall robustness.

STRUCTURE
Choose one of these:

- Strengths → Weaknesses (parallel bullet points)
- Strength → Validation → Limitation → Discussion
- Task-based critique (Problem 1, 2, 3)

LANGUAGE
Use O-Award style phrasing such as:

- “The accuracy depends heavily on the reliability of historical data...”
- “The computational complexity reflects a trade-off between efficiency and precision...”
- “The model remains idealized, but this abstraction supports rapid policy evaluation...”
- “For small-scale or extreme cases, uncertainty may increase...”

REQUIREMENTS

- Strengths must be supported by evidence (e.g., sensitivity analysis, R^2, average error, etc.).
- Weaknesses must be paired with a future improvement or mitigation.
- Mention computational complexity, data limitations, and generalizability.
- Reference uncertainty, thresholds, or scenario sensitivity when relevant.

User provides:

- Model names: <__填入你的内容_>
- What the model does well (results, metrics, plots): <_填入你的内容____>
- Known data or assumption limitations: <__填入你的内容_____>
- Computational constraints: <____填入你的内容_____>
- Future improvements you want to suggest: <____填入你的内容_____>

Write a professional O-Award level Strengths & Weaknesses section that:

- Sounds self-critical but confident
- Uses academic defense language
- Increases the perceived rigor of the model